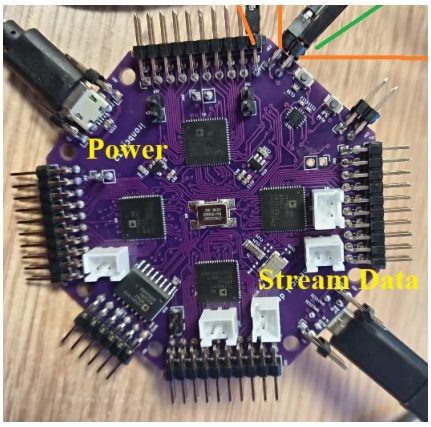
QUICK START & SAFETY GUIDE · SKU PIA24025P

From the Box to 32 Live Channels

Set up IronBCI-32 in seven steps: connect the electrodes, power the board from a USB battery, connect the data cable, find the serial port, stream and check the signal. Read the safety rules first.



WARNING Battery only, for everything.

IronBCI-32 has **no isolation circuitry**. Galvanic isolation from the mains depends entirely on the board **and** the laptop running from batteries. Power the board from a 5 V USB battery and run the laptop on its own battery, **unplugged from the wall**. IronBCI-32 is **not a medical device** and has not been certified by any regulator for use with the human body.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only, for everything.** Board on a 5 V USB battery; laptop on the data cable on its own battery, unplugged from the wall.
2	**No mains connection of any kind** while electrodes are on a person: no wall charger, no laptop charger, no desktop PC, no USB hub with its own power supply.
3	**No diagnosis.** Do not use IronBCI-32 to diagnose, monitor or treat any medical condition.
4	**Implanted devices.** Do not use it on a person with a pacemaker or other implanted electronic device without a physician's approval.
5	**Skin.** Use clean, intact skin. Stop if the skin becomes irritated, and clean off gel after each session.
6	**Power pins.** Never connect anything to the AVSS / AVDD pins at the end of each header.
7	**Exposed electronics.** The board has no case. Handle it by the edges, keep liquids away, or use a 3D-printed box.

Power while recording

Setup	While electrodes are on	Why
Board on a 5 V USB battery, laptop on battery	SAFE	No path to the mains.
Laptop plugged into its wall charger	NEVER	The data cable links the mains to the board and the person.
Board powered from a wall USB charger	NEVER	Connects the person to the mains.
Desktop computer on the data cable	NEVER	A desktop is always on mains power.

- Charge the USB battery and the laptop **before** the session, never during it.
- Check every cable before each session. A supervisor checks the setup when students use it.
- Clean electrodes with water after use and let them dry. Do not soak them in disinfectant or hot water.

02 WHAT YOU NEED

Item	Requirement
IronBCI-32	Included — 70 × 70 mm board, 2 × Micro-USB, 4 electrode headers
5 V USB battery	A USB power bank. Powers the board through its **Power** port.
2 × Micro-USB cables	One for power, one for data. The data cable must be a **full data cable** (4 wires), not charge-only.
Laptop	Windows, macOS or Linux with Python 3.10+, or Chrome / Edge. **Must run on battery** while recording.
Electrodes	Up to 32 dry or wet Ag/AgCl electrodes with 2.54 mm connectors, plus 2 ear-clip electrodes (REF, BIAS)
Cap and gel	32-channel cap (PiEEG 32-channel wet cap matches the layout) and conductive gel for wet electrodes

03 SETUP IN SEVEN STEPS

1 Gather the parts

IronBCI-32, a charged 5 V USB battery, two Micro-USB cables (one full data cable), a laptop on battery, 32 electrodes with two ear clips, and a cap.

2 Unplug the laptop

Disconnect the laptop from its wall charger and run it **on battery** for the whole session. Close other programs that may use serial ports.

3 Connect the electrodes

Fit the cap and plug the electrodes into the **bottom row** of the four headers. The reference and bias ear clips go on the **CH 1–8** header (REF: top row pin 1; BIAS: bottom row pin 2). See IRB32-PIN-004 for the pin order.

4 Connect the power cable

Plug the first Micro-USB cable from the 5 V USB battery into the port marked **Power**. The board starts and streams straight away — there is no on/off command.

5 Connect the data cable

Plug the second cable from the port marked **Data Stream** into the laptop. IronBCI-32 appears as a serial port; note its name.

System	How to find the port	Looks like
Windows	Device Manager → Ports (COM & LPT)	`COM3, COM6…`
Linux	`ls /dev/ttyACM*`	`/dev/ttyACM0`
macOS	`ls /dev/tty.usbmodem*`	`/dev/tty.usbmodem…`

6 Start streaming

PiEEG Server — dashboard at `http://localhost:1617`:

```
pip install "pieeg-server[ironbci32]"
pieeg-server --device ironbci32 --serial-port COM3          # Windows
pieeg-server --device ironbci32 --serial-port /dev/ttyACM0  # Linux / macOS
```

Browser — open cloud.pieeg.com in Chrome or Edge → PiEEG Devices → IronBCI-32 → pick the port. **BrainFlow** — see IRB32-SW-006.

7 Check the signal

Sit still, relax the face and run the quick tests below. If they behave as described, you are recording real biosignals.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All channels settle. Flat lines mean an electrode on the wrong row or not touching the skin.
Close the eyes for 10 s	Alpha (8–13 Hz) rises on the back-of-head channels (O1, Oz, O2, P3, Pz, P4), then drops when the eyes open.
Blink a few times	Large slow spikes on the front channels (Fp1, Fpz, Fp2).
Clench the jaw	Bursts of high-frequency noise, strongest on T7 and T8 — muscle (EMG), not brain activity.

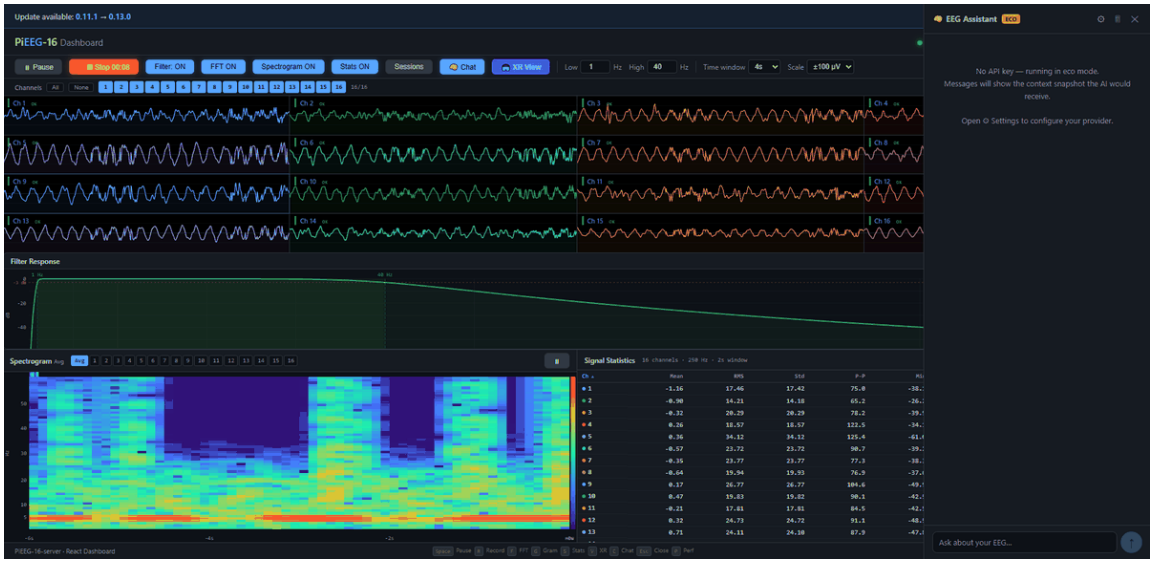


Figure 1 — PiEEG Server dashboard streaming 32 channels from IronBCI-32.

05 TROUBLESHOOTING

USB and serial port

Problem	Fix
No new serial port appears	The data cable is probably **charge-only**. Use a full data cable (4 wires) in the **Data Stream** port, not Power.
Port appears, but no data	The board also needs power on the **Power** port. Check the USB battery is on and charged.
Windows: unknown device	Unplug and replug the data cable. Windows 10 and 11 include the USB serial (CDC) driver.
Linux: `Permission denied`	`sudo usermod -aG dialout $USER`, then log out and in again (or run with sudo).
Port busy / Access is denied	Another program has the port open. Only one program can use it at a time.
Port name changes	Windows may assign a new COM number for a different USB socket. Check Device Manager each time.

Software

Problem	Fix
`BoardIds` has no `IRONBCI_32_BOARD`	BrainFlow is too old: `pip install -U brainflow` (tested with 5.20.1).
`IRONBBCI_32_BOARD` error in 2.All_ch.py	Typo in the script — change it to `IRONBCI_32_BOARD`.
`prepare_session` fails	Wrong port name. Set the serial port to the one you found.
PiEEG Server: pyserial missing	Install the extra: `"pieeg-server[ironbci32]"`.
Real-time plot does not open	`pip install pyqtgraph PyQt5`
PiEEG Cloud has no serial option	Use Chrome or Edge. Safari and Firefox do not support Web Serial.

Signal

Problem	Fix
Some channels completely flat	Electrode on the **top row** (shorted to REF). Move it to the bottom row.
Channel numbers look mixed up	Pins run 8-7-6-5-1-2-3-4 on each header, not 1–8. Check the pin order.
All channels noisy at once	Check the **REF and BIAS ear clips** first. Clean the earlobes and clip firmly.
Strong 50/60 Hz hum	The laptop is probably on its charger — **unplug it**. Move away from power strips and chargers.
Values in the tens of thousands of µV	Normal DC offset in raw data. Apply a band-pass filter, e.g. 1–40 Hz.
Slow spikes on front channels	Blinks and eye movements. Keep the eyes still or closed.
Bursts of high-frequency noise	Jaw, face or neck muscles. Relax the face; do not talk or chew.
Signal jumps when someone moves	Cable movement. Bundle the 32 leads, keep them short and still.

NOTE Board stopped working?

If no serial port appears with a known-good data cable and power, re-flash the firmware (IRB32-SW-006, section Firmware). Still stuck: Discord, GitHub issues or pieeg@pieeg.com.

06 RELATED DOCUMENTS

Doc. No.	Document	Contents
IRB32-DS-001	Product Datasheet	Features, specifications, block diagram, ordering
IRB32-MD-002	Mechanical & Dimensional Drawing	Outline, mounting holes, connector positions
IRB32-PCB-003	PCB Technical Information	Stack-up, layers, drill table, fabrication data
IRB32-PIN-004	Connector Pinout & Electrode Guide	Headers, pin order, REF/BIAS, 10-20 layout
IRB32-SW-006	Data Interface & Software Guide	Serial frame, µV conversion, SDKs, firmware